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1. How many times the following loop will execute?

```
int j = 3; while(j > 0) { cout << "Statements" << endl; j
    -= 2; }
```

1. **0**
2. **1**
3. **2**
4. **3**

2. In C++, a variable can be declared anywhere in the program this will increase _____.

1. **writability**
2. **readability**
3. **portability**

4. efficiency

3. In Flow Chart, flow of control is represented by _____.

1. **Rectangle**
2. **Circle**
3. **Diamomd**

4. Arrow

4. The only operator that the compiler overloads for user define data type by default is

1. **Plus (+) operator**
2. **MInus (-) operator**
3. **Assignment (=) operator**
4. **Equal (==) operator**

5. Once we have defined a symbolic constant value using
#define, that value _____ during program execution

1. **can be changed**
2. **cannot be changed**
3. **varies**
4. **becomes zero**

6. The function write() takes _____ as parameter(s).

1. **String of pointer type**

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2. **String of variable lengths, no. of bytes to be read and flags**
3. **Poimter array of characters and a delimiter**

4. String and no. of bytes to be written

7. _____ will be used for enclosing function statements into a block.

1. **" "**
2. **()**
3. **[]**

4. {}

8. What will be the result of the expression k = ++m; if initially k = 0 and m = 4?

1. **0**
2. **5**
3. **6**
4. **4**

9. What will be the correct syntax for initialization of a pointer ptr with string "programming"?

1. **char ptr = 'programming';**
2. **char *ptr = "programming";**
3. **char *ptr = 'programming';**
4. ***ptr = "programming";**

10. < and > both are _____ operators.

1. **Arithmetic**
2. **Relational**
3. **Logical**
4. **Mathematical**

11. _____ are not available in C language.

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1. **User defned functions**
2. **Built in functions**
3. **Library functions**
4. **Inline functions**

12. How many bytes an integer type pointer intPtr will jump in memory if the statement below is executed? intPtr += 2 ;

1. **2**
2. **4**
3. **8**
4. **12**

13. ostream is a _____ operator.

1. **dependent**
2. **member**
4. **None of the given**
3. **standalone**

14. _____ variables are those that are defined outside of main.

1. **Local**
2. **Dynamic**
3. **Globa l**
4. **Static**

15. _____ will return the number of bytes reserved for a variable or data type.

1. sizeof operator
2. **free operator**
3. **void pointer**
4. **new operator**

16. What will be the output of the following c++ code?

```
#include<iostream.h> #define max 100
main()
{
```

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```
#ifdef max
Cout<<"Hellow;
}
```

1. **Hello**
2. **"Hellow"**
3. **Max is 100**

4. Error

17. What will be the result of the expression k = ++m; if initially k = 0 and m = 5?

1. **0**
2. **5**
3. **6**
4. **4**

18. The memory address of the first element of an array is called _____.

1. **floor address**
2. **foundation address**
3. **first address**

4. base address

19. Symbolic constant PI can be defined as:

1. **#define PI 3.14 ;**
2. **#define PI 3.14**
3. **#define PI=3.14**
4. **# include pi= 3.14**

20. Body of any function is enclosed within _____.

1. **{ }**
2. **( )**
3. **[ ]**
4. **" "**

21. How many nested loops would be required to manipulate n-dimensional array?

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1. **n**
2. **n + 1**
3. **n - 1**
4. 2n

22. The loop which is most suitable to be used when the number of iterations is known is called _____.

1. for

2. while

3. **do-while**
4. **all looping processes require that the iterations be known.**

23. A pointer is _____.

1. **the address of a variable**
2. **an indication of the variable to be accessed next**
3. **a variable for storing address**
4. **the data type of an address variable**

24. _____ will explain the function of a program.

1. **Comments**
2. **Debugger**
3. **Compiler**
4. **Linker**

25. In C/C++, the default command line arguments passed to the main function are _____.

1. **float argc, char **argv**
2. **int argc, char **argv**
3. **int *argc, char *argv**
4. **int argc, float **argv**

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26. The memory allocation functions return a chunk of memory with a pointer of type _____.

1. **integer**
2. **float**
3. **ptr**
4. **void**

27. Encapsulation means _____.

1. **that the data of a class cannot be accessed from outside**
2. **that the data of a class can be accessed from outside**
3. **the data becomes public**
4. **that the data can be accessible anywhere within a main program**

28. A function is a block of statements that can be defined once and used _____ in the program.

1. **One time**
2. **Two times**
3. **Three times**

4. As many times as user wants

29. The malloc function takes _____ argument(s).

1. **two**
2. **three**
3. **four**

4. one

30. Return type of a function that does not return any value must be _____.

1. **char**
2. **int**
3. **voi d**
4. **double**

31. The parameter passed to isdigit() function is _____.

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1. **a character variable**
2. **a boolean variable**
3. **an integer variable**
4. **a character string**

32. Operating System is a type of a/an _____.

1. **application software**
2. **system softwar e**
3. **computer language**
4. **interpreter**

33. Array is a data structure that stores _____.

1. **Memory addresses**
2. **Variables**
3. **Data type**
4. **Data**

34. What will be the result of arithmetic expression 6+48/4*3?

1. **10**
2. **40.5**
3. **42**
4. **41**

35. In computer systems there are mainly _____ type of softwares.

1. **1**
2. **2**
3. **3**
4. **4**

36. The members of a class declared with the keyword struct are _____ by default.

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1. **static**
2. **Private**
3. **protected**
4. public

37. NULL has been defined in _____ header file.

1. **Iostream.h**
2. **Stdlib.h**
3. **Stdio.h**
4. **String.h**

38. Which of the function call is call by value for the following function prototype? float add(float);

1. **add(&x);**
2. add(x);
3. **add(float x);**
4. **add(*x);**

39. To access rand(), which library is required to be included in program?

1. **conio.h**
2. **stdio.h**
3. **stdlib.h**
4. **iostream.h**

40. _____ is used as a dereferencing operator.

1. *****
2. **+**
3. \-
4. **None of the above**

41. The name of the destructor is the same as that of a class proceeding with a _____.

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1. **& sign**
2. **# sign**
3. **@ sign**

4. ~ sign

42. _____ of a function is also known as signature of a function.

1. **Definition**
2. **Declaratio n**
3. **Calling**
4. **Invoking**

43. C++ is a _____ language.

1. **High level**
2. **Low level**
3. **Machine**
4. **Assembly language**

44. Which of the function call is "call by value" for the following function prototype?
float add(int);

1. **add(&x);**
2. **add(x) ;**
3. **add(int x);**
4. **add(*x);**

45. Function prototype is written,

1. **Within main function**
2. **After the return statement in main**
3. **Before the return statement in main**

4. Before call of that function

46. The members of a class declared without any keyword are _____ by default.

1. **protected**

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2. **private**
3. **public**
4. **constant**

47. _____ operator is used to pass the address of a variable in call by reference method.

1. **%**
2. **+**
3. **@**
4. &

48. Transpose of a matrix means that when we interchange rows and columns _____.

1. **the first row becomes the Last column**
2. **the first row becomes the first column**
3. **the Last row becomes the first column**
4. **the first column becomes the first row**

49. A 2D array multi[5][10] can be accessed using the array name as **multi, this technique is called _____.

1. **Single referencing**
2. **Single dereferencing**
3. **Double referencing**
4. Double dereferencing

50. What is the highest legal index for the following array?
int arr[4]

1. 4
2. 3
3. 2
4. 1

51. Within the statement obj1=obj2; obj1 will call the assignment operator function and obj2 will be passed as an argument to function.

1. True

2. False

52. The function seekg() takes _____ parameter(s).

1. 0 2. 1

3. 2

4. 3

53. The default mode for writing into a file using ofstream object is _____.

1. **out**
2. **bin**
3. **app**
4. **ate**

54. Which of the following operators is used to access the value of variable pointed by a pointer?

1. *** operator**
2. **-> operator**
3. **&& operator**
4. **& operator**

55. What will be the output of the given code?

```
#include #define MAX( A, B ) ((A) > (B) ? (A) : (B)) void main() {
int i, x, y; x = 23;
y = 45;
i = MAX( x++, y++ );
// Side-effect: // larger value incremented twice cout << "x = " << x << " y = " << y <<
'\n'; }
```

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1. **x=23 y=45**
2. **x=24 y=46**
3. **x=24 y=4 7**
4. **x=22 y=47**

56. An instance of a class is called _____.

1. **structure**
2. **data type**
3. **object**
4. **member function**

57. Constructor is a special function, called whenever we _____.

1. **create a function**
2. **instantiate an object of a class**
3. **destroy an object**
4. **create a class**

58. Which of the following is not a reserved word in C/C++?

1. **int**
2. **float**
3. **double**
4. sum

59. Once the _____ are created, they exist for the life time of the program.

1. **local variables**
2. **non static variables**
3. **static variables**
4. **automatic variables**

60. _____ executes all the lines before error and stops at the line which contains the error.

1. **Intrepreter**
2. **Compiler**
3. **Linker**
4. **Debugger**

61. If we have a program that writes the output data(numbers) to the disc, and if we collect the output data and write it on the disc in one write operation instead of writing the numbers one by one.

In the above situation the area where we will gather the number is called

1. **Heap**
2. **Stack**
3. **Buffe r**
4. **Cache**

62. What will be the output of following statement? cout<<setfill('0')<<setw(7)<< 128;

1. **0128128**
2. **000012 8**
3. **1280000**
4. **0012800**

63. What will be the output of following code segment?

```
for (int i = 2; i<10; i++) { if ( i == 5) continue;
cout << i << "," ;
}
```

1. **2 ,3,7,8, 9**
2. **2 ,3,4,6,7,8, 9**
3. **2 ,3, 4**
4. **4 ,6,7,8, 9**

64. Which of the following function call is "call by reference" for the following function prototype?

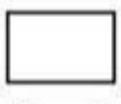
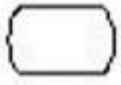
1. **func(int &num);**
2. **func(&num);**
3. **func(*num);**

4. **func(num);**

65. C is widely known as development language of _____ operating system.

1. **Windows**
2. **Unix**
3. **Mac OS**
4. **Linux**

66. Which one of the symbol is used to represent a decision in a flow chart?

1. 
2. 
3. 
4. 

67. Automatic variables are created on _____.

1. **Heap**
2. **Free store**
3. **Static storage**
4. stack

68. _____ statement is used to terminate the processing of a particular case and exit from switch structure.

1. **if**
2. **goto**

3. **break**
4. **continue**

69. Let suppose int a, b, c, d, e; a = b = c = d = e = 42;
This can be interpreted by the compiler as

1. **(a = b = (c = (d = (e = 42))));**
2. **a = (b = (c = (d = (e = 42))));**
3. **a = b = (c = (d = (e = 42)));**
4. **(a = b) = (c = d) = (e = 42);**

70. Word processor is a type of a/an _____.

1. **operating system**
2. application software

device driver utility
software

71. What is the output of the following code if the 3rd case is true

```
switch (var) { case 'a':cout<<"apple"<<endl; case
'b':cout<<"banana"<<endl; case
'm':cout<<"mango"<<endl; default: cout<<"any
fruit"<<endl; }
```

1. **mango**
2. **mang o**
   any frui t
3. **apple**
4. **None of the given**

72. How many dimensions does n-dimensional array has?

1. **n dimensions**
2. **2n dimensions**
3. **(n+1) dimensions**
4. **(n-1) dimensions**

73. dec, hex, oct are all _____.

1. **Member functions**
2. **Objects of input/output streams**
3. **Parameterized manipulators**

4. Non-parameterized manipulators

74. What is the output of the following code if the 2nd case is true

```
switch (var) { case 'a': cout<<"apple"<<endl; case
'b':cout<<"banana"<<endl; case
'm':cout<<"mango"<<endl; default: cout<<"any
fruit"<<endl; }
```

1. **banana**

2. **banana**

   any fruit

3. **banan a**

4. **None of the given**

   mango

   any fruit

75. Syntax of a union is identical to _____.

1. **structure**
2. **class**
3. **function**
4. **loop**

76. In order to get the right most digit of a number, we divide this number by 10 and take _____.

1. **Its remainder**
2. **Its quotient**
3. **Its divisor**
4. **The number**

77. Given a two dimensional array of integers, what would be the correct way of assigning the value 6 to the element at third row and fourth column?

1. **array[3][4] = 6 ; 2. array[2][4] = 6 ; 3. array[4][3] = 6 ;**

4. array[2][3] = 6 ;

78. Select the correct way to assign the address of first element of array to pointer?

1. int *ptr = &data[1];

2. int *ptr = &data;

int *ptr = data; int *ptr =

data[0];

79. In the given expression which operator will be evaluated first? 10 + (6 / 2) - 2 * 3?

1. **+**
2. **-**
3. **/**
4. *****

80. Whenever we use a library function or a predefined object or macro, we need to use a _____.

1. **source file**
2. **object file**
3. **header file**
4. **exe file**

81. C++ was developed by _____.

1. **Charles Babbage**
2. **Graham Bell**
3. **Bejarne Stroustrup**
4. **Von Nuemann**

82. A record is a group of related _____.

1. **Dat a**
2. **Field s**
3. **Bytes**
4. **Files**

83. !(x > 3) means in C++ that

1. **x is greater than 3**
2. **x is less than or equal to 3**
3. **x is less than 3**
4. **x is equal to 3**

84. What will be the result of the expression j = i++; if initially j = 0 and i = 5?

1. **0**

2. 5

3. 6

4. 4

85. A hierarchy of classes which are used to deal with console and disk files are called _____.

1. **Stream classes**
2. **Simple classes**
3. **Binary classes**
4. **IO classes**

86. getche() is a _____ function and defined in _____ header file.

1. **user-define function , conio.h**
2. **built-in function , conio.h**
3. **built-in function, stlib.h**
4. **built -in function, iostream.h**

87. _____ operators are the ones that require two operands on both sides of the operator.

1. **Double**
2. **Tow sided**

4. None of the given

3. Binary

88. How many elements are stored in the following? int matrix [4][5];

1. **9**

2. 20

3. 25

4. 10

89. The constructor contains _____.

1. **return type**

2. no return type

objects classes

90. _____ is a substitute of multiple if statement.

1. **if. . .elseif statement**
2. **Continue statement**
3. **Break statement**
4. **Default statement**

91. Constructor is itself a _____ of C++ and _____.

1. **class, can be overloaded**
2. **function, cannot be overloaded**
3. **function, can be overloaded**
4. **object, can not be initialized**

92. Which of the following is not an example of int data type?

1. **0**
2. **-32**
3. **65531**

4. -4

93. Overloaded assignment operator must be

1. **Member function of class**
2. **Non-member function of class**
3. **Friend function of class**
4. **Global function**

94. To read command-line arguments, the main() function itself must be given _____ arguments.

1. **1**

2. 2

3. 3

4. 4

95. How many bytes of memory are occupied by array 'str'?

char str[] = "programming";

1. **10**
2. **11**
3. **12**
4. **13**

96. How many times the following do-while loop will execute? int k = 10; do { cout << "Statements" << endl; k -= 2; } while(k>0);

1. **4**
2. **5**
3. **6**
4. **7**

97. if (a>b && a>c) then the condition will be true only if

1. **Both a>b and a>c are true**
2. **a>b is false and a>c is true**
3. **a>b is true and a>c is false**
4. **Both a>b and a>c are false**

98. Which function is used to locate the first occurance of a character in any string?

1. **strchr()**
2. **strstr()**
3. **strtok()**

4. strlen()

99. It is possible to return an object from function using this pointer.

1. **Tru e**
2. **False**

100. The _____ is called automatically when an object destroys

1. **destructor**
2. **constructor**
3. **main program**
4. **default constructor**

2nd section

Question 1

What is the other name of the variable?

- Identifier
- Visual
- Static
- Constant

Question 2

Which of the Following is true about streams?

A. It is a sequence of bytes
B. It is an ordered sequence
C. All bytes can go trough the stream simultaneously
D. Bytes that enters first into the stream will go out at last

- A &D
- A only
- C only
- A & B

Question 3

When No. of repetition is know then we use the ------ structure to perform repetition tasks.

- While loop
- For loop
- Switch
- Do while loop

Question 4

Shifting the binary number is similar to shifting the------ number.

- Base 7
- Decimal
- Hexadecimal
- octal

Question 5

We can read, write and manipulate the same file using ----------.

- Stdio.h
- Fsteram. h
- Iomanp.h
- Iostream.h

Question 6

In do while loop the , condition is checked ---------- one execution of loop.

- Before
- During
- At Any time
- After

Question 7

< , <=, >, >= are called ----- operators.

- Arithmetic
- Conational
- Logical
- Relational

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Question 8

For breaking complex problem into smaller pieces we use ----------.

- Function
- Loops
- If/else Break statement

Question 9

In three – dimensional array. We use------- 'for loop(s)' to populate the array.

- 3
- 2
- 1
- 4

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Question 10

a = a +1: can be written as ----------

- 1 + a = a:
- a+a+1
- a+=1
- a+1

Question 11

Continue statement usually force the immediate ------ iteration of the loop .

- last
- previous
- first
- next

Question 12

If a file is opened with ios: : out mode . then ----------.

- We can write in this file Content of file discarded
- A new file is created
- File is deleted

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Question 13

---------- operator is used to pass the address of variable in call by reference method.

- &
- @
- %
- +

Question 14

C++ is a ---------- language.

- Machine Low level Assembly language
- High level

Question 15

By writing a file seekg (15L. ios : : cur) we are moving ----------.

- 15 bit in the backward direction starting from the pervious position.
- 15 bit in the backward direction starting from the current position.
- 15 bytes in the backward direction starting from the pervious position.
- 15 bytes in the forward direction starting from the pervious position.

Question 16

Pointer points to a particular ----------.

- Data types
- Variable
- Memory address
- value

Question 17

If function has not been, declared before it is a -------.

- Run time error
- Logical error
- Syntax error

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- Late binding error

Question 18

---------- statement is used to terminate the processing of a particular case and exit from switch structure.

- Break
- If
- Goto continue

Question 19

Which one of the following is used to perform bit- wise OR operation?

- ~
- ^ ||
- |

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Question 20

In the following nested For Loop, which loop will run most number of times?

```
For ( int I = 0 : k : 5 K++)
{
    For (int k = 0 : k <: 5: k++)
{
……….
}
```

- Inner loop
- Depends upon the statements in the inner in the loop body
- Outer loop
- Both loop run equal number of time

Question 21

We can access a global variable ----------.

- From anywhere in the program
- From the main() function only

- From the function only
- From the loop only

Question 22

Structures are syntactically defined with the word----------.

- Struct
- Struc
- Structure
- None of the given

Question 23

If y= 20 and z = y++. Then the value of z will be ----------.

- 22
- 20
- 21
- 19

Question 24

The structure which can execute zero or more times is called ----------.

- For structure
- Do- while loop
- While structure
- None of the above

Question 25

Which of the following is the correct way to write a compound assignment operator?\

- X + = 4:
- X+=4:
- X+==4:
- None of the above

Question 26

Suppose we have a file with name “my file .txt”. What will be the correct syntax of file opening for the input.

- myFile. In(“myfile. txt”. Ios : :open):

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- myFile. open(“myfile. txt”. Ios : :in):
- myFile. Input(“myfile. txt”. Ios : :open):
- myFile. open(“myfile. txt”. Ios : :inpt):

Question 27

What will be the value of the variable output in the given piece of the code?

Double output = 0

Out put = (2 + 2)*4+ 2 /(4- 2):

- 17
- 11
- 15
- 12

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Question 28

Which one of the following languages has been used to write the compiler of “C ”language?

- Basic
- Fortran
- Java
- C

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Question 29

Char name []= “Hello world” :

In the above statement. A memory of ---------- characters will be allocated.

- 10
- 12
- 13
- 11

Question 30

What will be output of following code segment?

For (int I =2, i< 10: i++){

If (i==5)

Continue:

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Cout <<i<<".":

}

- 2, 3, 4
- 2, 3,4,6,7,8,9
- 4,6,7,8,9
- 2,3,7,8,9

Question 31

Identify the correct syntax for making a class friend of other class

- class Class one

{

friend otherclass

- friend ClassOne

{

OtherClass

- ClassOne

{

friend otherclass

- friend class ClassOne

{

friend otherclass

Question 32

The function will be return a reference to the global variable that exits throughout the promgram and thus there will be no danger of--------.

- Garbage collection
- Dangling reference

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- Wastage of memory
- System crash

Question 33

The friend function are ----------.

- Not member of class
- Member of class
- Inline function
- Void function

Question 34

The function of calloc takes two arguments, the first argument is the --------- and the second argument is the ----------.

- Required space in terms of number , size of space
- Size of space, number of bytes
- Space in terms of string character , size of list
- Size of array, number of bytes

Question 35

With the use of dynamic allocation of memory, the system resource can be----------.

- Wasted
- Used efficiently
- Used carelessly
- All of the above

Question 36

An instance of a class is called -------.

- Structure
- Data type
- Object
- Member function

Question 37

Initializing the data member within the class is ---------.

- A logical error
- A run time error

- A syntax error
- Not an error

Question 38

A friend function of a class is a function defined ----------.

- Inside that class and that has no access to the member of the class
- Output that class that right to access public member of the class only
- Outside that class and that has the right to access all member of class.
- Outside that class and that has the right to access all member of the class

Question 39

This reference to variable can be obtained by preceding the identifier of a variable with------.

- Dot operator
- Ampersand sign &
- ^sign *operator

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Question 40

In c language, the region of memory allocated at run time is called----------.

- Stack
- Heap
- Free store
- Available memory

Question 41

Constructor has ----------.

- No name
- The same of the class
- The same name as data member
- Return type

Question 42

Programmer should be very careful about the memory management because it can----------.

- Cause problem of memory leakage and dangling pointers.

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- Creates problem of null pointer
- Returns void pointers
- Affects the logic of the program

Question 43

Something after allocating memory we need additional space for this purpose we use

- Reallocation function
- Calloc function
- Realloc function
- Void pointer

Question 44

The constructor contains------.

- Return type
- No return type
- Objects
- Classes

Question 45

Encapsulation means----------.

- That the data of a class cannot be accessed from outside
- That the data of a class can be accessed from outside
- That data becomes public
- That the data can be accessed anywhere within a main program

Question 46

Friend classes are used in cases where one class is ---------- to another class

- Lightly coupled
- Loosely coupled
- Independent
- encapsulted

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Question 47

#define CIRCUMFERENCE(X) (2*pl * R) is a

- user defined function
- library function
- definition of a macro
- utility function

Question 48

Header files provide ---------- so the program running on one operating system can run without an error on the other system.

- Accessibility
- Reliability
- Accuracy
- portability

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Question 49

------ will return the number of bytes reserved for a variable or data type.

- Size of operation
- Free operator
- Void pointer New operator

Question 50

Default constructor takes----------.

- One parameter Two parameter
- No parameter
- Character type paremeter

Question 51

Void calloc(50,sizeroof(int))will return

- Char pointer
- A memory chunk of 50 integers from heap
- A memory chunk of 50 integers from stack
- A memory chunk with zero pointer

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Question 52

We cannot increment -------.

- Pointer
- Arrays
- References
- variables

Question 53

Symbolic constant pI can be defined as:

- #define PI 3.14;
- #define PI 3.14
- #define Pi =3.14
- #include pi= 3.14

Question 54

Special name which is substituted in code by its definition and as a reslt we get anexpanded code is called

- Union
- Directive
- Array
- Macro

Question 55

In a class we can have ------ constructor(s).

- Only one
- No Two
- Many

Question 56

C++ offers ---------- levels of data access control inside a class.

- Three
- Four
- Two
- five

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Question 57

For accessing data members we use ------- operator

- plus+
- multiplication*
- dot.
- Division/

Question 58

The memory allocation in C++ is carried out with the help of ----------.

- NULL Pointer
- New operator
- Dot operator
- +operator

Question 59

The object code of our program is combined with the ----------.

- Source program
- Machine code of the operating system
- Object code of the library functions
- With header files

Question 60

Initializing the data member with in the class is ----------.

- A logical error
- A run time error
- A syntax error
- Not an error

Question 61

Look at the program code and identify the error.

```
1  #include <iostream.h>
2  #define PI3.1415926;                    //Defining PI
```

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```
3  main()
{
4  int radius =5:
5  cout <<"Area of circle with radius " <<radius "="<<PI*radius * radius;
}
```

- It will be compiled successfully and will not generate error
- It will be compiled and return nothing
- Error exists in line number 2 semi colon is not allowed define directive
- The value of PI cannot be defined in the beginning

Question 62

What will be the output of the given code?

```
#include <iostearm.h>
Inline int max(int a, int b)
{
if(  a>b)
return  a;
return b;
}
main()
{
int I x,y;
x= 23; y=45
I = max (x++, y++);
Cout<< "x="<<"y=" '/n';
}
```

- X 23 y 45
- X 24 y 46
- X 22 y 46

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- X 23 y 47

Question 63

We should not use sch variable names that are starting with ---------- because in C++, there are lots of internal constant and symbolic names that star with it.

- Upper case alphabets
- Lower case alphabets
- Double underscore
- None of the given option

Question 64

For console input and output we use ----------.

- Conio.h header file
- Stdlib.h header file
- Process.h header file
- Getch.h header file

Question 65

The difference b/w pointer and references is that-------.

- We cannot do arithmetic with pointers
- We can do arithmetic with pointes
- We cannot reassign pointers
- We can assign references

Question 66

If you create a header file of your own and you save it in the “Header file” folder which is located on the current working directory then you must enclose your header file within----------.

- Braces
- Quotation marks
- #sign Angle brackets

Question 67

Reference Variables must----------.

- Not be initialized after they are declared
- Be initialized after they are declared
- Contain integer value
- Contain zero vlaue

Question 68

To prevent dangling reference the function returning reference should be sued with----------.

- Local variables
- Global variables only
- Arrays
- Static and global variables

Question 69

The function will return a reference to the global variable that exits throughout the program and thus there will be no danger of ----------.

- Garbage collection
- Dangling reference
- Wastage of memory
- System crash

Question 70

=operator is used for ----------.

- Checking equality
- Decision making
- Comparison
- Assigning values to cariables

Question 71

aFile- seeking (-10L,ios:cur)

- Both direction
- None of the given
- Backward direction
- Forward direction

Question 72

In C/C++, a variable's name can start with the symbol----------.

- -Underscore
- +Plus
- :Colon ;semicolon

Question 73

Which is the correct syntax to define a structure named "student"?

- Struct student(
  //variable
  );
- Struct student{
  //variable
  };
- Structure student(
  //variable
  );
- Structure student{
  //variable
  };

Question 74

The exclusive OR operator returns 1 if ----------.

- Both input are 1
- None of the given
- One in is 1 and other is zero
- Both input are 0

Question 75

If int a = 50; then the value of a/=3; will be

- 18
- 15
- 17
- 16

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Question 76

Identify the logical error in the following if condition

If ((i<5) &&(i>5))

- & will be used instead of &&
- && cannot be used with if condition
- Condition is always true
- Condition is always false

Question 77

Operating system is a type --------- software.

- Utility
- Device driver
- Application
- System

Question 78

MS word is type of ------- software. imp

- Application
- Device Driver
- Operating system
- utility

Question 79

What will be result of the expression

K=++m;

If initially K=0 and m=5?

- 6
- 4
- 0
- 5

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Question 80

Individual characters in a string stored in an array can be accessed directly using array--------.

- Subscript
- Script Value Superscript

Question 81

Structure use ---- for memory allocation

- Cache
- Stack
- Heap
- Queue

Question 82

a ^= b; can be written as

- a= b^a; a^=a+b
- a=a^+b;
- a=a^b;

Question 83

There is a pointer variable named ptr of type float. Which type of variable address can be stored by ptr?

- Int
- Char Float
- double

Question 84

Compiler translates high language program into------ language code.

- Java
- Machine
- C C++

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Question 85

From the following; which one is the correct syntax of an array declaration; array size is 5 and it offloat data type?

- [5] float name;
- Float name [5];
- Name[5]float;
- Float[5] name;

Question 86

The expression ((a +b)== 10) &&((X + Y)==20) is true only if----------.

- ((a+ b) == 10)is true
- Both ((a + b) ==10)and ((x +Y) == 20) are true
- Either ((a + b) == 10) or ((X +Y) == 20) is true
- ((X + Y) == 20 is true

Question 87

In do while loop the condition is checked --------- one execution of loop.

- During
- Before At any time
- After

Question 88

To get the value stored at a memory address , we use the----------.

- Referencing operator
- Binary operator
- dereferencing operator
- AND(&&) operator

Question 89

Following is the declaration of a ---------- array. Int arr[2][2][2]

- 2 –Dimensional
- 3 –Dimensional
- 1 –Dimensional

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- None of the given

Question 90

ofstream is used for _____.

A) Input file stream

B) Output file stream (100% Sure)

C) Input and output file stream

D) All of the given

Question 91

Base address is the memory address of _____ element of an array.

A) 1st (100% Sure)

B) 2nd

C) 3rd

D) 4th

Question 92

All elements of an array must be of _____ data type(s).

A) char and int

B) Same (100% Sure)

C) Different

D) float and double

Question 93 eof is used to check for the _____ of file when a file is being read.

A) Start

B) End (100% Sure)

C) Middle

D) Name

Question 93

While handling files, one can have _____ options.

A) read the file

B) write in some file

C) read and write in the same file

D) All of the given (100% Sure)

Question 94

Which of the following is true about streams?

A. It is a sequence of bytes

B. It is an ordered sequence

C. All bytes can go through the stream simultaneously

D. Bytes that enters first into the stream will go out at last

A) A only

B) C only

C) A and B (100% Sure)

D) A and D

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Question 95

we have opened a file stream myfile for reading(getting), myfile.tellg() gives us the current get position of the file pointer. It returns a whole number of type _____.

A) long (100% Sure)

B) int

C) short

D) double

Question 96

Null Character is represented by _____ in C++.

A) /0

B) \0 (100% Sure)

C) \n

D) \t

Question 97 suppose we
have int y[10];
To access the 4th element of the array we write.
A) y[4];

B) y[3]; (100% Sure)

C) y[2];

D) None of given

Question 98

What will be the size of the array declared as int array[1000];
A) 999

B) 1000

C) 1001 (100% Sure)

D) 1002

Question 99

_____ Returns true if c is a letter and false otherwise.
A) int isalpha(int c) (100% Sure)

B) int isalnum(int c)

C) int isxdigit(int c)

D) int isdigit(int c)

Question 100

What will be the correct syntax of declaration of the following statement?
"ptr is a pointer to const int"
A) int const* ptr

B) int * const ptr

C) const int * ptr (100% Sure)

D) const * int ptr

Question 101

What will be the size of following array? int arr[29];

A) 0

B) 30 (100% Sure)

C) 29

D) 28

Question 102

Individual characters in a string stored in an array can be accessed directly using array _____.

A) superscript

B) script

C) subscript (100% Sure)

D) value

Question 103 char **argv can be read as _____.

A) pointer to pointer

B) pointer to char

C) pointer to pointer to char (100% Sure)

D) None of the given

Question 104

By default, the starting index of an array in C++ is _____ .

A) 2

B) -1

C) 0 (100% Sure)

D) 1

Question 105

The ASCII code of null character is _____.

A) 000 (100% Sure)

B) 010

C) 111

D) 110

Question 106

To manipulate n-dimensional array _____ nested loops are required.

A) n-2

B) n-1

C) n+1

D) n (100% Sure)

Question 107 dereferencing operator is represented by _____.

A) * (100% Sure)

B) +

C) _

D) None of the given

Question 108

Given a two dimensional array of integers, what would be the correct way of assigning the value 6 to the element at third row and fourth column?

A) array[3][4]=6 (100% Sure)

B) array[2][4]=6

C) array[4][3]=6

D) array[2][3]=6

Question 109

Which of the following is true about streams?

A. It is a sequence of bytes

B. It is an ordered sequence

C. All bytes can go through the stream simultaneously

D. Bytes that enters first into the stream will go out at last

A) A only

B) C only

C) A and B (100% Sure)

D) A and D

Question 110

_____ Returns true if c is a letter and false otherwise.

A) int isalpha(int c) (100% Sure)

B) int isalnum(int c)

C) int isxdigit(int c)

D) int isdigit(int c)

Question 111

Individual characters in a string stored in an array can be accessed directly using array _____.

A) superscript

B) script

C) subscript (100% Sure)

D) value

Question 112

we have opened a file stream myfile for reading(getting), myfile.tellg() gives us the current get position of the file pointer. It returns a whole number of type _____.

A) long (100% Sure)

B) int

C) short

D) double

Question 113

From the following; hich one is the correct syntax of an array declaration: array size 5 and it is of float data type?

A) float [5] name;

B) name [5] float;

C) float name[5]; (100% Sure)

D) None of the given options

Question 114

Pointers are a special type of _____ in which a memory address is stored.

A) variable (100% Sure)

B) Location

C) Characters

D) None of the given

Question 115

_____ is the pointer which determines the position in a file from where the next read operation occurs.

A) Set

B) Get

C) Put

D) Tell (100% Sure)

Question 116

We use _____ to open a file or stream for extraction.

A) in (100% Sure)

B) out

C) app

D) ate

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Question 117

All elements of an array must be of _____ data type(s).

A) char and int

B) Same (100% Sure)

C) Different

D) float and double

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Question 118

Suppose that an integer type pointer contains a memory address 0x22f220. What will be the new memory address if we increment this pointer by one?

A) 0x22f221

B) 0x22f222

C) 0x22f223 D) 0x22f224 (100% Sure) Question 119

In C++, what will be the highest index of the following array?

int array[15];

A) 16 (100% Sure)

B) 15

C) 14

D) 13

Question 120

To get the value stored at a memory address, we use the _____.

A) refrencing operator

B) dereferencing operator (100% Sure)

C) simple operator

D) None of the given

Question 121

_____ are conventional names of command line parameters of the 'main()' function.

A) 'argb' and 'argv'

B) 'argc' and 'argv' (100% Sure)

C) 'argc' and 'argu'

D) None of the given

Question 122

In Analysis, we try to have a _____.

A) Determined the inputs

B) Break up of problem

C) Precise problem statement (100% Sure)

D) None of the given

Question 123

These functions seekg() and seekp() requires an argument of type _____ to let them how many bytes to move forward or backward.

A) int

B) short

C) long (100% Sure)

D) double

Question 124 char name []
="Hello World";
In the above statement, a memory of _____ charaters will be allocated.
A) 13

B) 11

C) 12 (100% Sure)

D) 10

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Question 125

Individual characters in a string stored in an array can be accessed directly using array _____.
A) superscript

B) script

C) subscript (100% Sure)

D) value

Question 126

While handling files, one can have _____ options.
A) read the file

B) write in some file

C) read and write in the same file

D) All of the given (100% Sure)

Question 127

The increment of a pointer depends on its _____.

A) variable

B) value

C) data type (100% Sure)

D) None of the given

Question 128

All elements of an array must be of _____ data type(s).

A) char and int

B) Same (100% Sure)

C) Different

D) float and double

Question 129

_____ Returns true if c is a digit and false otherwise.

A) int isalpha(int c)

B) int isalnum(int c)

C) int isxdigit(int c)

D) int isdigit(int c) (100% Sure)

Question 130

What will be the size of the following character array?

char name[]="Adeel";

A) 5

B) 6 (100% Sure)

C) 4

D) 7

Question 131

While handling files, one can have _____ options.

A) read the file

B) write in some file

C) read and write in the same file

D) All of the given (100% Sure)

Question 132

In Analysis, we try to have a _____.

A) Determined the inputs

B) Break up of problem

C) Precise problem statement (100% Sure)

D) None of the given

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Question 133

By default, the starting index of an array in C++ is _____.

A) 2

B) -1

C) 0 (100% Sure)

D) 1

Question 134

The string in the array is terminated by a _____.

A) zero

B) nil

C) null (100% Sure)

D) one

Question 135

The statement cout << yptr will show the _____ the yptr points to.

A) Value

B) memory address (100% Sure)

C) Variable

D) None of the given

Question 136

When we access a multidimensional array, each index is surrounded by _____.

A) Single quotes ''

B) Double quotes ""

C) Brackets [] (100% Sure)

D) Commas ,,

Question 137

There is a pointer variable named ptr of type float then address of which type of variable the ptr will store in it?

A) int

B) float (100% Sure)

C) char

D) double

Question 138

Null character is represented by _____ in C++.

A) /0

B) \0 (100% Sure)

C) \n

D) \t

Question 139

What will be the size of the array declared as
int array[1000];

A) 999

B) 1000

C) 1001 (100% Sure)

D) 1002

Question 140

Suppose that an integer type contains a memory address 0x22f220. What will be the new memory address if we increment this pointer by one?

A) 0x22f221

B) 0x22f222

C) 0x22f223

D) 0x22f224 (100% Sure)

Question 141

We use _____ to open a file or stream for extraction.

A) in (100% Sure)

B) out

C) app

D) ate

Question 142

Whenever some number is added in an array name, it will jump as many _____ as the added number.

A) rows (100% Sure)

B) value

C) column

D) None of the given

Question 143

From following; which one is the correct syntax of an array initialize:
Array size is 10 and it is of double data type to value 0?

A) arr[10]={0.0};

B) double arr[10]=0.0;

C) double arr[10]={0.0}; (100% Sure)

D) double arr[]=0.0;

Question 144

An object of _____ class can be created of saving the current position of any opened file before reading from the file.

A) Stream

B) Filepos

C) Streampos (100% Sure)

D) Filestream

Question 145

What will be the size of the following character array?
char name[]="Adeel";

A) 5

B) 6 (100% Sure)

C) 4

D) 7

Question 146

We can say that a pointer is a variable that hold the _____ address.

A) byte code

B) source code

C) memory (100% Sure)

D) compiler

Question 147

By default, the starting index of an array in C++ is _____.

A) 2

B) -1

C) 0 (100% Sure)

D) 1

Question 148

Given a two dimensional array of integer, what would be the correct way of assigning the value 6 to the element at third row and fourth column?

A) array[3][4]=6; (100% Sure)

B) array[2][4]=6;

C) array[4][3]=6;

D) array[2][3]=6;

Question 149

The parameter passed to isdigit() function is _____ variable.

A) Character

B) Boolean

C) Integer (100% Sure)

D) Float

Question 150

We must include the header file _____ to convert the value of one type into another type using built-in functions.

A) conio.h

B) stdlib.h (100% Sure)

C) string.h

D) iostream.h

Question 151

A record is a group of related _____.

A) Data

B) Fields (100% Sure)

C) Bytes

D) Files

Question 1502

A _____ is an array of characters that can store number of character specified.

A) Char

B) String (100% Sure)

C) Multidimensional array

D) Data type

Question 153

To manipulate n-dimensional array _____ nested loops are required.

A) n-2

B) n-1

C) n+1

D) n (100% Sure)

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Question 154

How we declare an array of integer data whose size is 20 with array name "count".

A) int count[19];

B) int count[20]; (100% Sure)

C) float count[20];

D) int count[21];

Question 155

_____ contains function for manipulations of character data.

A) ctype.h (100% Sure)

B) iostream.h

C) string.h

D) None of the given

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Question 156

The statement cout<< yptr will show the _____ yptr points to.

A) Value

B) memory address (100% Sure)

C) Variable

D) None of the given

Question 157

By default, the starting index of an array in C++ is _____.

A) 2

B) -1

C) 0 (100% Sure)

D) 1

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Question 159

Which header file must be included to use the functions tolower() an toupper()? A) iostream.h

B) conio.h

C) ctype.h (100% Sure)

D) stdlib.h

Question 160

To manipulate n-dimensional array _____ nested loops are required.

A) n-2

B) n-1

C) n+1

D) n (100% Sure)

Question 161

When we access a multidimensional array, each array index is surrounded by _____.

A) Single quotes "

B) Double quotes

C) Brackets [] (100% Sure)

D) Commas ,,

Question 162

There is a pointer variable named ptr of type float then address of which type of variable the ptr will store in it?

A) int

B) float (100% Sure)

C) char

D) double

Question 163

We have opened a file stream myfile for reading (getting), myfile.tellg() gives us the current get position of the file pointer. It returns a whole number of type _____.

A) long (100% Sure)

B) int

C) short

D) double

Question 163

Transpose of a matrix means that when we interchange rows and columns _____.

A) the first row becomes the Last column

B) the first row becomes the first column (100% Sure)

C) the Last row becomes the first column

D) the first column becomes the first row

Question 164 dereferencing operator is represented by _____.

A) * (100% Sure)

B) +

C) -

D) None of the given

Question 165

To get the value stored at a memory address, we use the _____.

A) referencing operator

B) dereferencing operator (100% Sure)

C) simple operator

D) None of the given

Question 166 eof is used to check for the _____ of file when a file is being read.

A) Start

B) End (100% Sure)

C) Middle

D) Name

Question 167

We can define a matrix as _____ array.

A) Sorted

B) Unsorted

C) Single dimensional (100% Sure)

D) Multi dimensional

Question 168

Which header file must be included to use the functions tolower() an toupper()? A) iostream.h

B) conio.h

C) ctype.h (100% Sure)

D) stdlib.h

Question 169

aFile.seekg(-10L, ios:cur) indicates the file pointer will move 10 bytes in the _____.

A) forward direction

B) backward direction (100% Sure)

C) both direction

D) None of the above

Question 170

In _____, we try to have a precise problem statement.

A) Analysis (100% Sure)

B) Design

C) Coding

D) None of the given

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Question 171

If an array has 50 elements what is allowable range of subscripts?

A) 0-49 (100% Sure)

B) 1-49

C) 0-50

D) 1-50

Question 172

At the desing phase, we try to break up the problem into _____.

A) functional units (100% Sure)

B) non-functional units

C) small units

D) None of the given

Question 173

The name of the array is a constant pointer which contains the memory address of the _____ of the array.

A) first element (100% Sure)

B) last element

C) second element

D) none of the given

Question 174

"multi [3][3]" element can also be accessed by '_____'.

A) **(multi+3)+3

B) *(*(multi+3)+3) (100% Sure)

C) *(multi+3)

D) None of the given

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Question 175

_____ function give the position of the next character to be read from that file.

A) tellp()

B) tellg() (100% Sure)

C) seekg()

D) seekp()

Question 176

These functions seekg() and seekp() requires an argument of type _____ to let them how many bytes to move forward or backward.

A) int

B) short

C) long (100% Sure)

D) double

Question 177

How many bytes an integer type pointer intPtr will jump in memory if the statement below is executed?
intPtr += 2 ;

A) 2

B) 4

C) 8 (100% Sure)

D) 12

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Question 178

In C-language the string "AZMAT" is equal to _____.

A) Azmat

B) azmat

C) AZMAT (100% Sure)

D) None of the given

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Midterm short Notes

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1. Describe History of C++?

The C++ programming language has a history going back to 1979, when Bjarne Stroustrup was doing work for his Ph.D. thesis. One of the languages Stroustrup had the opportunity to work with was a language called Simula, which as the name implies is a language primarily designed for simulations.

2. What is Dev. C++?

Bloodshed Dev-C++ is a full-featured Integrated Development Environment (IDE) for the C/C++ programming language. It uses Mingw port of GCC (GNU Compiler Collection) as it's compiler. Dev-C++ can also be used in combination with Cygwin or any other GCC based compiler.

3. Why Use Dev. C++ instead of C++ DOS IDE?

Dev-C++ is a free IDE for Windows that uses either MinGW or TDM-GCC as underlying compiler. Originally released by Bloodshed Software, but abandoned in 2006, it has recently been forked by Orwell, including a choice of more recent compilers.

4. What is #Include directive or Copy books or Header File?

Many programming languages and other computer files have a directive, often called include (as well as copy and import), that causes the contents of a second file to be inserted into the original file. These included files are called copybooks or header files.

5. Why we use curly braces in c?

Curly braces are used to group a set of statements. Often we use them along with loops and conditional statements in order to avoid confusion and to define a clear scope.

6. What is Main () function?

In C, the "main" function is treated the same as every function, it has a return type (and in some cases accepts inputs via parameters). The only difference is that the main function is "called" by the operating system when the user runs the program.

7. How many data types are in C?

Void, Int, Float, Character, String

8. How many types are for integers?

Int or signed int unsigned int, short int or signed short int, long int or signed long int, unsigned long int

9. How many types are for Float data type?

Float, double, long double

10. What is mean by using namespace std; in C?

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The built in C++ library routines are kept in the standard namespace. That includes stuff like cout, cin, string, vector, map, etc. Because these tools are used so commonly, it's popular to add "using namespace std" at the top of your source code so that you won't have to type the std:: prefix constantly.

11. Define the memory size of int variable types? Type Size in Bytes Range int or signed int

2 -32,768 to 32767 unsigned int 2 0 to 65535 short int or signed short int 1

-128 to 127 long int or signed long int 4 -2,147,483,648 to 2,147,483,647

unsigned long int 4 0 to 4,294,967,295

12. Define the memory size of Float Variable types?

Type Size(bytes) Range Float 4

3.4E-38 to 3.4E+38 double 8 1.7E-308 to

1.7E+308 long double 10 3.4E-4932 to

1.1E+4932

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13. How declare a variable?

The Syntax of Declare variable is data_type variable_name; where the data_type is the type of data which can hold a variable variable_name might be anything except the C directives and reserve words.

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14. What is namespace?

Consider a situation, when we have two persons with the same name, Zara, in the same class. Whenever we need to differentiate them definitely we would have to use some additional information along with their name, like either the area if they live in different area or their mother or father name, etc. Same situation can arise in your C++ applications. For example, you might be writing some code that has a function called xyz() and there is another library available which is also having same function xyz(). Now the compiler has no way of knowing which version of xyz() function you are referring to within your code.A namespace is designed to overcome this difficulty and is used as additional information to differentiate similar functions, classes, variables etc. with the same name available in different libraries. Using namespace, you can define the context in which names are defined. In essence, a namespace defines a scope.

15. How define a namespace?

A namespace definition begins with the keyword namespace followed by the namespace name as follows

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```
namespacenamespace_name {

  // code declarations

}
```

To call the namespace-enabled version of either function or variable, prepend the namespace name as follows:

```
name::code;  // code could be variable or function.
```

16. What are reserve words in C?

Reserve or Keywords are the words that C language use for his personal use a user cannot use reserve words as variable, function or class name. Example: IF, Else, Switch, For, While, Float etc.

17. What rules are for declaring a variable?

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1. Every variable name should start with alphabets or underscore (_).
2. No spaces are allowed in variable declaration.
3. Except underscore (_) no other special symbol are allowed in the middle of the variable declaration.
4. Maximum length of variable is 8 characters depend on compiler and operation system.
5. Every variable name always should exist in the left hand side of assignment operator.
6. No keyword should access variable name.

18. What is difference between declaration and initialization of a Variable?

For a variable, a definition is a declaration which allocates storage for that variable.

Initialization is the specification of the initial value to be stored in an object, which is not necessarily the same as the first time you explicitly assign a value to it.

19. What is cin and cout?

cout is used for output, cin for input. Cout and cin are not key words in the C++ language. They are variables, instances of classes that have been declared in <iostream>.

20. Define local and global variable?

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Variables that are declared inside a function or block are local variables. They can be used only by statements that are inside that function or block of code. Local variables are not known to functions outside their own.Global variables are defined outside of all the functions, usually on top of the program. The global variables will hold their value throughout the life-time of your program. A global variable can be accessed by any function. That is, a global variable is available for use throughout your entire program after its declaration.

21. Define Syntax Error?

Asyntax error is an error in the syntax of a sequence of characters or tokens that is intended to be written in a particular programming language. For compiled languages, syntax errors are detected at compile-time. A program will not compile until all syntax errors are corrected.

22. What is Run-Time Error?

An error that occurs during the execution of a program. In contrast, compile-time errors occur while a program is being compiled. Runtime errors indicate bugs in the program or problems that the designers had anticipated but could do nothing about. For example, running out of memory will often cause a runtime error.

23. What is Logical Error?

A logic error is a bug in a program that causes it to operate incorrectly, but not to terminate abnormally (or crash). A logic error produces unintended or undesired output or other behavior, although it may not immediately be recognized as such.

24. What are arithmetic operators?

C language has 5 arithmetic operators (+, -, *, /, %). Arithmetic operators are used to perform arithmetic operations in c programming.

25. Define Logical Operators?

C Language have 3 types of Logical operator or conditional operators.

Operator	Description	Example
&&	Called Logical AND operator. If both the operands are non-zero, then the condition becomes true.	(A && B) is false.
||	Called Logical OR Operator. If any of the two operands is non-zero, then the condition becomes true.	(A || B) is true.
!	Called Logical NOT Operator. It is used to reverse the logical state of its operand. If a condition is true, then Logical NOT operator will make it false.	! (A && B) is true.

26. What is Conditional operator?

A ternary operator or Conditional Operator is an operator that takes three arguments. The arguments and result can be of different types. Many programming languages that use C-like syntax feature a ternary operator, ?:, which defines a conditional expression. Since this operator is often the only existing

ternary operator in the language, it is sometimes simply referred to as "the ternary operator". In some languages, this operator is referred to as "the conditional operator". Another example for a ternary operator is between, as used in SQL and Python.

27. Difference between operators and operands?

Operands are the objects that are manipulated and operators are the symbols that represent specific actions. For example, in the expression. 5 + x. and 5 are operands and + is an operator. All expressions have at least one operand.

28. Describe about file streams? Data Type Description

ofstream This data type represents the output file stream and is used to create files and to write information to files.

ifstream This data type represents the input file stream and is used to read information from files.

fstream This data type represents the file stream generally, and has the capabilities of both ofstream and ifstream which means it can create files, write information to files, and read information from files.

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29. What are conditional statements?

In computer science, conditional statements, conditional expressions and conditional constructs are features of a programming language, which perform different computations or actions depending on whether a programmer-specified Boolean condition evaluates to true or false. Example: IF. Then, Switch.

30. What are repletion statements?

In C programming Language it is often necessary to repeat the same block of code a given number of times, or until a certain condition is met. Executes a sequence of statements multiple times and abbreviates the code that manages the loop variable. You can use one or more loops inside any other while, for, or do. While loop.

31. Define Nested If and Nested Loop?

In C++ we can use if statement in the else block. Or we can also include if block in another if block, it's called nested IF. The placing of one loop inside the body of another loop is called nesting. When you "nest" two loops, the outer loop takes control of the number of complete repetitions of the inner loop. While all types of loops may be nested, the most commonly nested loops are for loops.

32. What is pre-processor Directive?

The C preprocessor modifies a source file before handing it over to the compiler, allowing conditional compilation with #ifdef, defining constants with #define, including header files with #include.

33. Why use strcpy() function?

This function use to copy string from one string variable to other string variable. Example strcpy(s1, s2) copy string from s2 to s1.

34. What is a Function?

A function is a group of statements that together perform a task. Every C program has at least one function, which is main (), and all the most trivial programs can define additional functions. You can divide up your code into separate functions.

35. What are parameters of a function?

In computer programming, a parameter is a special kind of variable, used in a subroutine to refer to one of the pieces of data provided as input to the subroutine. These pieces of data are called arguments. An ordered list of parameters is usually included in the definition of a subroutine, so that, each time the subroutine is called, its arguments for that call can be assigned to the corresponding parameters.

36. What is a pointer?

The variable that stores the address of another variable is called a pointer. Pointers are a very powerful feature of the language that has many uses in lower level programming.

37. Difference between a pointer variable and a Variable?

Both are variables a pointer hold the memory address of a variable and an Variable that hold his value both variable must be same type to work with each other.

38. What is a Class?

A class in C++ is a user defined type or data structure declared with keyword class that has data and functions (also called methods) as its members whose access is governed by the three access specifies private, protected or public (by default access to members of a class is private).

39. What is a Class object?

Object is a class type variable. Objects are also called instance of the class. Each object contains all members (variables and functions) declared in the class. We can access any data member or member function from object of that class using. Operator.

40. How can we refer to the global variable if the local and the global variable names are same?

We can apply scope resolution operator :: for the scope of global variable. The scope resolution operator helps to identify and specify the context to which an identifier refers, particularly by specifying a namespace. The specific uses vary across different programming languages with the notions of scoping. In many languages the scope resolution operator is written "::".

41. What is recursion?

Recursion is the process of repeating items in a self-similar way. In programming languages, if a program allows you to call a function inside the same function, then it is called a recursive call of the function.

42. What is an identifier?

An Identifier can only have alphanumeric characters (a-z, A-Z, 0-9) and underscore (_). The first character of an identifier can only contain alphabet (a-z, A-Z) or underscore (_). Identifiers are also case sensitive in C. For example name and Name are two different identifier in C.

43. What is friend class and friend Function?

Friend Function:A C++ friend functions are special functions which can access the private members of a class. They are considered to be a loophole in the Object Oriented Programming concepts, but logical use of them can make them useful in certain cases.

Friend Class: A C++ friend class is a special class that can access the members, objects and methods of another class.

44. What are constructors and destructors?

Constructor: A class constructor is a special member function of a class that is executed whenever we create new objects of that class. A constructor will have exact same name as the class and it does not have any return type at all, not even void. Constructors can be very useful for setting initial values for certain member variables.

Destructors: A destructor is a special member function of a class that is executed whenever an object of its class goes out of scope or whenever the delete expression is applied to a pointer to the object of that class.A destructor will have exact same name as the class prefixed with a tilde (~) and it can neither return a value nor can it take any parameters. Destructor can be very useful for releasing resources before coming out of the program like closing files, releasing memories etc.

45. What is the difference between actual and formal parameters?

The parameters sent to the function at calling end are called as actual parameters while at the

Receiving of the function definition called as formal parameters.

46. What is difference between including the header file with-in angular braces <> and double quotes “”?

If a header file is included with in <> then the compiler searches for the particular header file

Only with in the built in include path. If a header file is included with in “ “, then the compiler

Searches for the particular header file first in the current working directory, if not found then in the built in include path

47. If a pointer is declared for a class, which operator can be used to access it?

Arrow operator (->) is used for accessing members of structure using pointer variable, whenever we declare structure variable then member can be accessed using the dot operator. But when pointer to a structure is used then arrow operator is used. Both dot and arrow operator serves same function to access member of structure.

48. What is function overriding?

Function overloading (also method overloading) is a programming concept that allows programmers to define two or more functions with the same name. Each function has a unique signature (or header), which is derived from: function/procedure name. Number of arguments. Arguments' type.

49. Is it possible to have Virtual Constructor? If yes, how? If not, Why not possible?

There is nothing like Virtual Constructor. The Constructor can't be virtual as the constructor

Is a code which is responsible for creating an instance of a class and it can't be delegated to any other object by virtual keyword mean.

50. What is a dangling pointer?

A dangling pointer arises when you use the address of an object after its lifetime is over. This may occur in situations like returning addresses of the automatic variables from a function orusing the address of the memory block after it is freed.

51. What is a smart pointer?

A smart pointer is a C++ class that mimics a regular pointer in syntax and some semantics, but it does more. Because smart pointers to different types of objects tend to have a lot of code in common, almost all good-quality smart pointers in existence are template by the pointe type.

52. What is the difference between a pointer and a reference?

A reference must always refer to some object and, therefore, must always be initialized; pointers do not have such restrictions. A pointer can be reassigned to point to different objectswhile a reference always refers to an object with which it was initialized.

53. What is the difference between const char *myPointer and char *const myPointer?

Const char *myPointer is a non-constant pointer to constant data; while char *const myPointer is a constant pointer to non-constant data.

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